

Meeting 6

Lex Janssens

Date:	17 June 2025
Subject:	How Stable are Baselines: analysis on the Consistency of Reinforcement Learning Baselines
Mentor:	dr.ir. Á. Serra-Gomez
Project term:	February 2025 – August, 2025
E-mail:	l.p.janssens@umail.leidenuniv.nl

Outcome of previous meeting

- Extend research to a third algorithm: TD3
- Extend research to a third framework: TorchRL
- Design plots showing the relative ranking of algorithms using a single set of hyperparameters among different frameworks and environments.
- Start writing, beginning with preliminaries, and background.

Results during the last weeks

- Performed an experiment on the effects of vectorization of the environment: using the same algorithm (SAC) and the same hyperparameters, CleanRL shows significant differences in performance between $n_{envs} = 1$ and $n_{envs} = 8$ due to the inconsistent definition of 'a timestep'.
- TD3 shows some inconsistencies, specifically with the use of Stable Baselines 3 hyperparameters. Thus far, I found that Stable Baselines 3 target actor works with normalized actions, whilst that isn't the case for CleanRL.
- The relative ranking of algorithms across a set of environments, where the hyperparameters of these algorithms are Stable Baselines 3 defaults, is inconsistent among frameworks. Interestingly enough, this relative ranking is still preserved when using CleanRL default hyperparameters or original paper hyperparameters, suggesting that Stable Baselines 3 hyperparameters and implementational details (whether it is (hyper)parameter constants or methods) have synergy.
- Started with a third framework: TorchRL. Tianshou, as the first choice, had a cumbersome installation process on ALICE. I am working on a first test run, where TorchRL uses hyperparameters of other frameworks on HalfCheetah.
- Written several pages of preliminaries and background + a figure 1. Halted explaining the algorithms until I am certain of TorchRL's implementational details (or drop it).

Plans for next weeks

- Drop TorchRL? → It takes quite a while, not only to make sure TorchRL runs properly, but also to analyze the algorithms. It is a nice addition, but in my opinion, it is better to focus on writing.
- Finish analysis TD3
- Finishing up running experiments, continue writing, and find causalities for the results within frameworks.
- First draft? → End of July would be a safe date given aforementioned planning.